

System Documentation: Selective VPS Configuration Backup

Pull-Method via WireGuard Tunnel

Author: Jacek Kaplon | linkedin.com/in/jacek-kaplon

1. Architecture & Design Principles

- **Model:** Pull Backup — Proxmox initiates the connection and fetches data from the VPS.
- **Transport:** Encrypted WireGuard tunnel (point-to-point).
- **Authentication:** SSH key-based (Ed25519) on non-standard port **50022**.
- **Efficiency:** Incremental backups using hardlinks (**--link-dest**).
- **Storage:** Local mount point **/mnt/usb-backup/** on Proxmox.

2. Network Configuration (WireGuard)

Side A: VPS (Hub)

File: /etc/wireguard/wg-backup-vps.conf

```
[Interface]
Address = 10.50.50.1/24
ListenPort = 51855
PrivateKey = <VPS_PRIVATE_KEY>
MTU = 1380

[Peer]
# Proxmox (rasputin)
PublicKey = 6IHnWuMmm0YmS09d6FmSJxrYIW4MVYV2lehoK9Fv/ww=
AllowedIPs = 10.50.50.2/32
```

Side B: Proxmox (Client / Receiver)

File: /etc/wireguard/wg-bkp-pve.conf

```
[Interface]
Address = 10.50.50.2/24
PrivateKey = <PROXMOX_PRIVATE_KEY>
MTU = 1380

[Peer]
PublicKey = AFLRKUYyG++CTFffcrCeRTL2kVMXJdGxde7tpLhGxA0=
Endpoint = <VPS_PUBLIC_IP>:51855
AllowedIPs = 10.50.50.1/32
PersistentKeepalive = 25
```

3. SSH Key Authorisation (Over VPN)

For rsync to operate without a password prompt, the Proxmox public key must be present on the VPS.

Step 1 — Generate key pair (on Proxmox):

```
ssh-keygen -t ed25519 -f /root/.ssh/id_ed25519 -N ""
```

Step 2 — Copy public key to VPS (port 50022):

```
ssh-copy-id -p 50022 -i /root/.ssh/id_ed25519.pub root@10.50.50.1
```

4. Backup Script (Proxmox)

File: /root/backup-vps-configs.sh (remember to run `chmod +x`)

```
#!/bin/bash
# --- CONFIGURATION ---
DATE=$(date +%F_%H-%M)
REMOTE_USER="root"
REMOTE_HOST="10.50.50.1"
REMOTE_PORT="50022"
LOCAL_BASE="/mnt/usb-backup/vps-configs"
LOCAL_TARGET="$LOCAL_BASE/$DATE"
LOCAL_LATEST="$LOCAL_BASE/latest"
LIST_FILE="/root/vps-configs-list.txt"
SSH_KEY="/root/.ssh/id_ed25519"

# Create base directory
mkdir -p "$LOCAL_BASE"

echo "[1/3] Starting incremental backup from VPS..."

# Main rsync command
rsync -aAXv \
    --delete \
    --link-dest="$LOCAL_LATEST" \
    --files-from="$LIST_FILE" \
    -e "ssh -p $REMOTE_PORT -i $SSH_KEY" \
    $REMOTE_USER@$REMOTE_HOST:/ "$LOCAL_TARGET"

# Update the 'latest' symlink
echo "[2/3] Updating latest symlink..."
ln -sf "$LOCAL_TARGET" "$LOCAL_LATEST"

echo "[3/3] Backup complete: $LOCAL_TARGET"
```

Filter File (File List)

File: /root/vps-configs-list.txt

```
etc/wireguard/
etc/ssh/sshd_config
etc/nginx/
etc/letsencrypt/
var/spool/cron/crontabs/
etc/prometheus/
etc/grafana/
```

5. Schedule (Crontab)

Edit with crontab -e on Proxmox:

```
# 1. Run backup every day at 03:00 AM
00 3 * * * /bin/bash /root/backup-vps-configs.sh >> /var/log/vps-backup.log 2>&1

# 2. Delete backups older than 7 days every day at 03:30 AM
30 3 * * * find /mnt/usb-backup/vps-configs -maxdepth 1 -type d -name "202*" -mtime +7 -exec rm -rf {} \;
```

6. Troubleshooting

Problem	Solution
Ping works but SSH hangs	Verify that MTU=1380 is set on both sides of the tunnel.
rsync: command not found	Run <code>apt install rsync</code> on both servers.
Permission denied (publickey)	Check that the port in the script (-p 50022) matches the SSH port configured on the VPS.
link_stat failed (2)	A directory from the list does not yet exist on the VPS — safe to ignore.